Semantic Retrieval and Live-Data Reranking for Personalized Travel Recommendation

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Abstract

Travel destination selection requires travelers to reconcile personal preferences with practical factors including budget, climate, transportation, lodging, safety, and changing travel conditions. This paper presents an AI-assisted travel decision-support system that transforms natural-language travel requests into ranked destination recommendations. The recommender uses a two-stage pipeline in which explicit geographic and trip-type constraints are applied before semantic retrieval and preliminary deterministic scoring. The highest-ranked candidates are then enriched with live travel information and five-year historical seasonal climate evidence before a second deterministic scoring stage. Final recommendations must satisfy live-data viability requirements for both flight and lodging information. Semantic retrieval uses asymmetric Jina query and passage embeddings with cosine similarity, while a persistent embedding cache reduces repeated embedding requests. Gemini is used only after ranking to generate natural-language explanations from structured recommendation evidence and does not determine destination order. The current destination inventory contains 199 records across 17 fields. A scenario-based evaluation comprising nine completed travel requests produced 44 final recommendations and showed strong handling of explicit numeric budgets, dates, positive travel categories, and supported geographic constraints. The evaluation also exposed limitations involving negative preferences, priority language, lodging geographic validation, accommodation classification, missing advisory states, and extreme flight itineraries. These findings characterize the system as an implemented and inspectable recommendation architecture rather than a trained ranking model or an empirically optimal recommender.

Keywords

travel recommendation, recommender systems, semantic retrieval, cosine similarity, decision support, contextual recommendation, large language models

1 Introduction

Destination selection is a multi-criteria decision problem in which travelers must reconcile personal preferences with feasible constraints such as budget, trip duration, climate, transportation, lodging, safety, and changing destination conditions. Travel recommender systems attempt to reduce this decision burden by identifying alternatives that more suitably match traveler requirements. Existing tourism recommender research includes collaborative, content-based, context-aware, knowledge-based, and mixed approaches, with the choice of method depending on the available user, destination, and interaction data [1]. Context can also materially affect recommendations; prior tourism research has incorporated variables such as temperature, precipitation, season, location, and traveler characteristics when selecting destinations [2].

The system was developed as a request-centered travel recommender that does not require a history of user ratings or previous trips. A request specifies an origin and free-form travel preferences and may additionally provide an explicit budget. The application extracts structured constraints from the prompt, filters incompatible destinations, measures semantic similarity between the request and candidate destinations, and computes an initial ranking from locally available information. A limited shortlist then proceeds to more expensive contextual evaluation using live flight and lodging data, five-year historical seasonal climate, and country-level travel-advisory information.

The recommendation mechanism deliberately separates candidate retrieval from final decision scoring. Semantic similarity helps determine which destinations receive deeper evaluation, but it does not directly determine final ordering. Shortlisted candidates are reevaluated through an explicit multi-factor scoring procedure incorporating trip-type compatibility, seasonal climate evidence, budget compatibility, stored safety and attraction values, and advisory penalties. Final results must also satisfy viability requirements that require usable live flight and lodging evidence.

Generative AI is used after the deterministic ranking is complete. Gemini receives the ordered recommendation records and supporting evidence and is instructed to explain rather than rerank the destinations. A deterministic explanation path remains available if generative output fails. This separation prevents the language model from acting as the ranking authority.

The contribution of the project is therefore architectural and integrative rather than the introduction of a newly trained recommendation algorithm. The system combines rule-based constraint extraction, geographic and categorical filtering, asymmetric semantic retrieval, deterministic multi-criteria scoring, selective live-data enrichment, viability filtering, destination replenishment, and evidence-grounded explanation in one recommendation workflow. The evaluation further examines which types of constraints are handled reliably and where implementation-level validation is still necessary.

2 Literature Review

2.1 Tourism Recommendation Approaches

Tourism recommender systems address information-selection problems in which travelers choose among destinations, attractions, accommodations, activities, and related alternatives. Badouch and Boutaounte describe personalized tourism recommendation through

collaborative filtering, content-based filtering, and mixed approaches while identifying destination, accommodation, attraction, and activity recommendation as distinct tourism applications [1]. Their review also identifies contextual factors such as climate, budget, activity preferences, seasonality, and travel restrictions, illustrating that destination suitability can depend on several criteria rather than a single similarity measure.

The information available to a recommender influences which approach is appropriate. Collaborative methods commonly depend on interactions between users and items, whereas content-based approaches can operate using characteristics associated with candidate items. Yoon and Choi similarly distinguish collaborative and content-based tourism recommendation and discuss the difficulty of interaction-dependent methods when information about new users or destinations is unavailable [2]. The system considered in this paper does not maintain historical user-rating or user-destination interaction data. It instead interprets each request independently and compares the request with available destination information.

More computing-intensive tourism methods have also been investigated. Huang proposes a personalized travel-route model using neural representations, self-attention, user information, geographic information, and social-interaction data to learn traveler preferences [3]. Kiritbhai and Ranpara compare several AI-based recommender approaches for travel search, including collaborative filtering, content-based methods, hybrid systems, matrix factorization, and deep learning, and emphasize tradeoffs involving accuracy, scalability, diversity, and interpretability [7]. That literature provides useful context for algorithm decision, but the present application does not train a ranking network over historical user behavior. Pretrained embeddings are used as a semantic representation service, while the final ranking remains deterministic.

2.2 Context-Aware and Multi-Criteria Recommendation

Destination suitability can change with the context in which a recommendation is requested. Yoon and Choi's R2Tour system incorporates situational variables including temperature, precipitation, season, and location together with traveler-profile information [2]. Their work indicates that environmental context can participate directly in tourism recommendation rather than being displayed only as supplemental information.

The present recommender addresses a related problem through a different methodology. It does not train the classifiers used by R2Tour. Instead, seasonal climate information is converted into explicit scoring adjustments after candidate shortlisting. The literature therefore motivates the use of contextual variables without implying that the two systems implement the same model.

Travel recommendation can also be formulated as a multi-criteria decision problem. Hernawati et al. combine the Analytical Hierarchy Process (AHP) with Simple Additive Weighting (SAW) to rank tourism alternatives using multiple weighted criteria [4]. Beraldi et al. similarly present a tourism decision-support architecture that combines learned preference scores with routing optimization while respecting user-specified restrictions such as budget and time windows [5]. These studies provide a relevant basis for treating travel recommendation as constrained decision support rather than similarity matching alone. The present implementation does not use AHP, SAW, or routing optimization; its weights are developer-defined heuristic parameters rather than formally derived criterion weights.

2.3 Recommendation Evidence and Explanation

A numerical recommendation score alone does not communicate which competing factors produced the result. This is especially relevant when destinations differ across cost, weather, transportation, safety, and attraction characteristics. The implementation retains the factors used during scoring and exposes them to a separate explanation component.

This design should not be interpreted as proof of human-centered explainability or increased user trust. Rong et al.'s survey of human-centered explainable AI emphasizes that explanation quality is commonly evaluated through human-centered properties such as trust, understanding, usability, and human–AI collaboration, and also notes that human evaluation remains comparatively sparse [6]. Those outcomes therefore require dedicated user evaluation. The implemented contribution is narrower: numerical scores, contextual observations, pricing information, advisory status, and human-readable scoring reasons remain inspectable, while the language model communicates an already-determined ranking rather than choosing destinations itself.

2.4 Positioning of the Proposed System

The reviewed research demonstrates several directions in tourism recommendation, including interaction-driven personalization, content-oriented recommendation, context-aware modeling, learned travel-route prediction, constrained decision support, and explicit multi-criteria decision methods [1–5, 7]. The proposed application does not attempt to reproduce all of these approaches. It addresses a request-centered setting in which a traveler provides natural-language requirements without requiring an established behavioral profile.

3 Methodology

The recommender implements a two-stage destination-selection pipeline that separates inexpensive candidate screening from more costly contextual evaluation. A request passes through validation, constraint extraction, hard filtering, semantic retrieval, preliminary scoring, live-data enrichment, final scoring, viability filtering, optional destination replenishment, and explanation generation.

3.1 Request Validation and Constraint Extraction

The primary recommendation endpoint accepts a starting destination, a natural-language travel prompt, and an optional numeric budget. FastAPI exposes the endpoint, while Pydantic validates the request structure before recommendation processing begins.

The prompt is converted into structured values using deterministic parsing functions. Budget extraction recognizes dollar-denominated amounts and common budget expressions. An explicit API budget takes precedence over a value extracted from the prompt; when neither is present, the system uses a default budget of $2,500.

Additional parsing functions determine trip duration and travel month. Explicit month names are mapped directly to calendar months, while recognized seasonal expressions can be mapped to representative months. These values are used to construct outbound and return dates and to obtain seasonal historical weather information.

Travel categories and geographic constraints are also extracted from controlled keyword and mapping tables. The application can identify supported trip types, countries, broad regions, subregions, and selected administrative areas. When multiple geographic levels are detected, more detailed constraints take precedence rather than broader ones.

3.2 Candidate Eligibility Filtering

Before semantic retrieval, the destination inventory is filtered using explicit requirements recognized by the parser. The traveler's origin is removed from the candidate set, and supported geographic constraints are applied at the most specific available level.

Explicitly recognized trip categories can also restrict eligibility. A destination remains eligible when its stored travel-style labels intersect the requested categories. These hard filters reduce the candidate set before semantic similarity or live contextual information is considered.

3.3 Semantic Retrieval

The application uses the Jina `jina-embeddings-v3` model to add semantic relevance to candidate selection. User requests are encoded using the `retrieval.query` task, while destination representations are encoded using `retrieval.passage`. Each destination representation contains geographic information, climate classification, safety and attraction attributes, trip-type information, and descriptive travel text.

Let the user-query embedding be represented by vector $\mathbf{q}$ and a destination embedding by $\mathbf{d}$. Cosine similarity is calculated as

$$\cos(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\|_2 \|\mathbf{d}\|_2}. \tag{1}$$

If either vector has zero magnitude, a neutral value is returned. If embedding retrieval fails or a usable embedding is unavailable, candidates receive a semantic contribution of zero and the deterministic recommendation path continues.

Destination embeddings are persisted in `destination_embeddings.json`. Cache keys are derived from a SHA-256 hash of the embedding-model identifier and destination text; therefore, changing either input produces a new cache entry. At the time of the implementation audit, the cache contained 393 vectors with 1,024 dimensions. The cache count exceeds the active destination count because entries derived from earlier destination-text representations may remain.

3.4 Preliminary Candidate Scoring

Performing live flight, lodging, climate, and advisory queries across the full inventory would require unnecessary external-service calls. The application therefore calculates a preliminary score from locally available information before live enrichment.

The local score includes signals derived from trip-type compatibility, geographic match, climate category, locally estimated cost, stored safety, and attraction quality. A region-based round-trip flight estimate and stored nightly lodging estimate provide an inexpensive initial cost estimate. No live travel-pricing request is required at this stage.

Semantic similarity is then added to the deterministic score. For destination d,

$$S_{\text{pre}}(d) = S_{\text{local}}(d) + 60 \max\left(\cos(\mathbf{q}, \mathbf{d}), 0\right), \tag{2}$$

where $S_{\text{local}}(d)$ is the preliminary deterministic score. Negative cosine values do not create a negative semantic penalty.

Eligible destinations are ordered by S_{pre}. The highest-ranked eight candidates proceed to live contextual evaluation during an evaluation round. This constitutes a candidate-generation and reranking design rather than a single monolithic scoring process.

3.5 Contextual Travel Enrichment

3.5.1 Historical Seasonal Climate. The active recommendation path uses historical seasonal climate rather than present-day weather to evaluate future travel months. When a travel month can be resolved, the application queries the Open-Meteo Archive using destination coordinates and calculates a five-year historical monthly mean from the five most recent complete calendar years.

For each destination and requested month, the system derives average high temperature, average low temperature, an overall monthly temperature estimate, and precipitation information. These values are compared with broad warm, mild, or cold preferences detected in the request. The resulting values are historical meteorological evidence and should not be interpreted as a forecast for the future travel dates.

Current-weather functionality exists elsewhere in the application, but it is not used by the active final destination-scoring path evaluated in this paper.

3.5.2 Flight and Lodging Pricing. Concrete trip dates are generated before live pricing is requested. When an explicit month is available, the outbound date is placed on the fifteenth day of the next occurrence of that month; the return date is then calculated from the resolved trip duration.

Live flight pricing is requested through SerpAPI's Google Flights interface using origin and destination airport codes. Returned flight records can include total fare, airlines, itinerary durations, layovers, connection counts, airport codes, travel dates, and a Google Flights search link.

Lodging retrieval uses SerpAPI's Google Hotels interface. The service evaluates candidate properties using rating, review count, property information, price, and availability of a user-facing link. The application attempts to exclude hostels and generally requires a rating of at least 3.5 and at least 20 reviews. The least expensive qualifying property is used as the Best Value lodging component for trip pricing. Additional quality and resort properties can also be presented but do not determine the Best Value trip total.

The trip-cost estimate is

$$C_{\text{trip}} = C_{\text{flight}} + C_{\text{lodging}}. \tag{3}$$

The logistics layer can calculate deterministic fallback values when live retrieval fails. However, the current final-recommendation

viability rules require usable live flight and Best Value lodging evidence before a destination can remain in the final result set.

3.5.3 Travel Advisories. Country-level travel advice is retrieved from GOV.UK Foreign Travel Advice and mapped into four internal advisory levels. These internal levels are used only by the application and should not be interpreted as a numbering system published directly by GOV.UK.

Advisory status is converted into a fixed penalty:

$$P_{\text{adv}} = \begin{cases} 0, & L = 1, \\ -5, & L = 2, \\ -15, & L = 3, \\ -30, & L = 4. \end{cases} \tag{4}$$

Domestic United States destinations are handled separately. When a usable advisory record is not available, the pipeline can continue with an unavailable advisory state rather than aborting the entire request.

3.6 Final Destination Scoring

After contextual enrichment, a new deterministic score is calculated rather than treating the preliminary score as the final decision value. Semantic similarity therefore influences which candidates receive expensive evaluation but does not directly determine final order.

For shortlisted destination d, the final score can be represented as

$$S_{\text{final}}(d) = T(d) + W_h(d) + B(d) + S(d) + A(d) + P_{\text{adv}}(d), \tag{5}$$

where T is the final trip-type contribution, W_h is the historical-climate contribution, B is the budget contribution, S and A are stored safety and attraction values, and P_{adv} is the advisory penalty.

A recognized trip-type match contributes 30 points. A trip estimate within the resolved budget contributes 25 points, while an estimate exceeding budget receives a 20-point deduction. Stored safety and attraction values are added directly. Climate adjustments depend on the requested weather category and observed historical monthly values.

The final scoring function is therefore a heuristic model rather than a learned or probabilistically calibrated ranking function.

3.7 Final Viability and Candidate Replenishment

A high final score alone is insufficient for recommendation. Each candidate must also satisfy live-data viability requirements. The current implementation requires usable nonfallback flight information from SerpAPI, including a price, airline information, and a usable search link. Best Value lodging must likewise be backed by a nonfallback SerpAPI result with a selected property, usable total price, and link.

Let $V(d) \in \{0, 1\}$ indicate whether destination d satisfies these live-data requirements. The final set can be expressed as

$$R = \text{Top5}\left(\{d \mid V(d) = 1\}, S_{\text{final}}(d)\right). \tag{6}$$

The target is five viable recommendations. If too few candidates survive an evaluation round, the application can request additional destination discovery and rerun candidate evaluation. The current implementation permits up to three replenishment rounds and can request up to six newly discovered destinations during a replenishment step.

This behavior means that the eight-candidate shortlist is a per-round live-evaluation limit rather than an absolute maximum number of candidates that can be considered during an entire recommendation request.

Table 1: Destination Inventory Schema

Field	Role
`city`	Destination city identifier
`country`	Destination country
`region`	Broad geographic region
`subregion`	More specific geographic grouping
`administrative_area`	State/province or equivalent area
`trip_type_1`	Primary travel category
`trip_type_2`	Secondary travel category
`trip_type_3`	Tertiary travel category
`climate_tag`	Categorical climate label
`average_humidity`	Optional stored humidity field
`estimated_nightly_cost`	Local lodging fallback estimate
`safety_score`	Stored safety value
`attraction_score`	Stored attraction value
`latitude`	Geographic latitude
`longitude`	Geographic longitude
`airport_code`	Airport identifier
`destination_description`	Textual destination summary

3.8 Fundamental Explanation Generation

The deterministic ranking is completed before Gemini is invoked. Ranked recommendation records are serialized with destination order, scores, trip costs, flight and lodging information, climate evidence, safety and attraction values, advisory information, fallback indicators, and scoring reasons.

Gemini is instructed to preserve the official ordering and explain supplied evidence rather than choose or rerank destinations. Hard factual fields such as prices, dates, links, and ranking positions are rendered from structured data. If generative output is unavailable or incomplete, a deterministic explanation can be produced from the recommendation records.

4 Datasets and Data Sources

The application uses a hybrid data architecture consisting of a persistent destination inventory and request-time contextual information retrieved from external services. The local inventory establishes the candidate space and stores fields required for filtering, preliminary ranking, semantic retrieval, and fallback calculations. External services provide information that can change over time.

4.1 Destination Inventory

The current `travel_locations.csv` contains 199 destination records and 17 fields. No duplicate city-country pairs were identified during the implementation audit. Table 1 summarizes the schema.

Table 2: External Services Used by the Application

Service	Role
Jina AI	Query and destination embeddings
Geoapify	Geocoding, discovery, and location enrichment
Open-Meteo	Five-year historical seasonal climate
SerpAPI Google Flights	Live flight search and pricing
SerpAPI Google Hotels	Live lodging search and pricing
GOV.UK	Country-level travel-advisory information
Gemini	Evidence-grounded explanation generation and selected fallback classification tasks
OpenWeather	Current-weather functionality outside the active final-ranking path

The inventory functions as a persistent candidate collection rather than a conventional supervised-learning dataset. It contains no target class and is not divided into training and testing partitions.

At audit time, climate labels comprised 108 warm, 79 mild, and 12 cold destinations. The stored nightly-cost field ranged from $90 to $230, with a mean of approximately $149.75 and a median of $150. The `average_humidity` field was empty across all 199 records, while `administrative_area` was missing for 147 of the 199 destinations. These missing values were retained in the evaluated dataset rather than imputed late in the project, so the manuscript and reported scenario results describe the same dataset state used during testing. These missing values limit the role those fields can play in the current pipeline.

4.2 Destination Discovery, Validation, and Persistence

The destination inventory is mutable. Geoapify can be used to discover and geocode additional destinations when the current candidate set is insufficient. New records can be normalized, enriched, validated, and written to the CSV for subsequent requests.

Validation checks include nonempty city and country values, valid coordinate ranges, a three-letter airport code when one is supplied, positive estimated lodging cost, and bounded safety and attraction values. Duplicate city-country records are checked before persistence.

Newly discovered records can initially receive default heuristic values for estimated nightly cost, safety, and attraction. At the same time, additional metadata such as trip types, climate classification, and destination description are enriched separately. The precise provenance of the earliest manually stored safety, attraction, and fallback-cost values could not be reconstructed during manuscript preparation. These attributes are therefore treated as prototype ranking inputs rather than externally validated measurements.

4.3 External Contextual Sources

Table 2 summarizes the primary external services used by the implementation.

Dynamic flight and lodging information is not permanently written as destination characteristics. Seasonal climate values are calculated during recommendation processing from archived observations. Advisory information is likewise retrieved at request time and contributes to final scoring without becoming a static destination field.

4.4 Data Limitations

Several properties of the data architecture affect interpretation. Stored safety and attraction values and local fallback cost estimates are heuristic attributes rather than continuously refreshed measurements, and their original provenance could not be fully reconstructed. The destination inventory, while broader than the original seed set, is not an exhaustive representation of all travel destinations. Missing `administrative_area` and `average_humidity` values were documented rather than backfilled after evaluation, avoiding a dataset change that could make the final manuscript inconsistent with the tested system state.

Historical weather is based on a five-year monthly mean. It therefore provides seasonal context rather than a forecast. Live flight, lodging, and advisory information can change after an evaluation run. Exact recommendation outputs are consequently time-dependent.

5 System Implementation

The application uses a Streamlit frontend and FastAPI backend with separate service modules for recommendation orchestration, destination management, semantic embeddings, travel coordination, flight retrieval, lodging retrieval, advisory retrieval, geographic enrichment, region resolution, caching, and explanation generation.

The frontend collects the origin, free-form prompt, and budget, starts a recommendation job, polls job progress, and renders the final result. The backend maintains recommendation jobs in an in-process dictionary protected by a lock. This design supports asynchronous interaction for the prototype but does not provide durable job maintenance across backend restarts.

The destination inventory is loaded with an in-memory cache, while embeddings are persisted separately in JSON. A general TTL cache service also exists in the repository, but it is not consistently integrated into all external-service calls; therefore, the manuscript does not claim that every API response is actively cached.

The service-driven organization supports failure isolation. Embedding failure can reduce semantic similarity to a neutral value, explanation failure can invoke deterministic prose, and fallback travel estimates can be calculated internally. However, final recommendation viability in the current implementation requires evidence of successful live flight and lodging.

6 Evaluation and Results

The system was evaluated using a scenario-based end-to-end test suite designed to exercise different stages of the recommendation pipeline. Because the project does not contain a labeled ground-truth destination dataset, historical user-rating corpus, or formal user study, conventional recommender-system metrics such as precision, recall, and ranking accuracy cannot be calculated without introducing artificial relevance labels. The evaluation therefore focuses on observable system activity, constraint fulfillment, and the validity of the supporting travel information returned with each recommendation.

Nine scenarios were executed using Boston as the trip origin. The prompts varied travel style, geographic restrictions, season, duration, budget, safety emphasis, positive and negative preferences, and flight preferences. Each resulting recommendation was inspected for destination relevance, geographic compliance, budget compliance, date consistency, seasonal-climate evidence, travel-advisory handling, flight availability, lodging availability, and fallback use. Across the nine completed scenarios, the system produced 44 final destination recommendations.

Flight and lodging records were additionally reviewed for practical validity. Flight records were inspected for origin, destination, dates, connection count, layover duration, and unusually long itineraries. Lodging results were checked for conformity with the intended destination and for cases in which accommodation type conflicted with the intended lodging filters. Successful retrieval from an external service was therefore not treated automatically as proof that the result was appropriate.

Table 3 summarizes the executed scenarios.

6.1 Budget and Date Handling

Budget constraints were among the most consistently handled inputs. In Scenario S7, the traveler stated that total cost could not exceed $1,000 and that remaining under budget was more important than every other preference. All five displayed recommendations remained below that ceiling: San Juan at $620, Virginia Beach at $717, Myrtle Beach at $722, Hilton Head Island at $892, and Cancun at $554.

The same pattern appeared across the other evaluated scenarios: no displayed final recommendation exceeded the explicit numeric budget supplied for its scenario. This does not guarantee that live prices remain bookable at the observed values because travel prices are dynamic, but it demonstrates consistent observed handling of the resolved numeric budget.

Trip-duration and date handling were also consistent in the observed outputs. Five-night requests produced five-night lodging intervals, six-night requests produced six-night intervals, and structured flight and hotel dates matched the generated itinerary windows.

6.2 Geographic Constraint Management

Explicit geographic restrictions recognized by the parser performed strongly. Scenario S4 restricted the destination set to Florida and returned Key West, Miami, Fort Lauderdale, and Naples, all satisfying the state-level condition.

Scenario S8 requested destinations only in Japan and explicitly prohibited destinations outside the country. The returned set consisted of Kyoto, Okinawa, Hokkaido, Osaka, and Nara. Scenario S9 similarly restricted recommendations to the Caribbean and returned San Juan, Montego Bay, Aruba, Punta Cana, and Nassau.

These observations support the narrower claim that geographic expressions represented by the parser can operate effectively as candidate filters. They do not establish that arbitrary geographic language outside the supported vocabulary will always be resolved correctly.

6.3 Positive and Negative Preference Handling

A substantial difference was observed between positive preferences and negative constraints. Positive requests such as beach, culture, history, nature, and broad climate preferences were frequently reflected in destination metadata and scoring reasons.

Scenario S5 was designed to test exclusionary language. The request asked for a warm destination with beaches and nature but explicitly excluded nightlife and major cities. Key West nevertheless ranked first even though nightlife appeared in its destination metadata. Miami ranked second despite being both a major city and a destination whose stored travel categories include city and nightlife.

The generated explanation acknowledged some of these conflicts after ranking. This indicates that the language-generation stage can describe a mismatch that the deterministic constraint and ranking stages did not prevent. The observed implementation was therefore more reliable at identifying positive requested attributes than at translating exclusions into hard candidate constraints.

6.4 Preference Priority and Safety Handling

Safety-related information demonstrably contributes to scoring. In Scenario S7, Cancun received an internal Level 3 advisory and the implemented 15-point penalty. Destinations with Level 1 advisory records received no advisory deduction.

Scenario S6 tested the phrase "safety is my highest priority." Kyoto had the strongest static safety score among the returned destinations, 9/10, and also had a Level 1 advisory, yet ranked fifth. This does not imply that Kyoto should necessarily rank first because other factors, including travel practicality, matter. It does show that natural-language emphasis such as "highest priority" does not automatically convert safety into a dominant final-ranking weight.

Scenario S9 exposed a related issue involving unavailable advisory evidence. The prompt prohibited destinations with an advisory above Level 1, but San Juan remained eligible when the advisory service returned no advisory level. In the current scoring logic, unavailable advisory evidence can therefore have the same numerical effect as a verified no-penalty advisory state. For hard safety constraints, unknown evidence should be distinguished from verified Level 1 status.

6.5 External Lodging Validation

The most significant non-parser errors occurred during lodging validation. Two unambiguous geographic mismatches were observed. In Scenario S2, Jackson, Wyoming was paired with lodging associated with Jackson/Ridgeland, Mississippi. In Scenario S9, the Aruba recommendation used Wingate by Wyndham Memphis East as its Best Value lodging.

These cases demonstrate that a usable provider response does not guarantee geographic correctness. A stronger validation rule would require both provider usability and geographic consistency:

$$V_{\text{property}} = V_{\text{provider}} \wedge V_{\text{geography}}. \tag{7}$$

Accommodation classification also produced repeated inconsistencies. Hostel-style properties such as Juliette in San Juan, Beehive

Table 3: Scenario-Based Evaluation of the Recommendation System

ID	Primary Test Condition	Outcome	Principal Finding
S1	Warm beach trip in March, five nights, total budget of $2,500	Pass	Five beach-oriented destinations were returned within budget, with live travel information and five-year climate evidence.
S2	Cold-weather nature trip in January	Pass with issue	Cold-weather and nature-oriented destinations were retrieved, but a Jackson, Wyoming recommendation was paired with lodging in Mississippi.
S3	Europe-only cultural city trip in October	Pass	All five destinations were in Europe and aligned with the requested city/cultural characteristics, dates, and budget.
S4	Florida-only trip emphasizing warm weather, beaches, food, and nightlife	Pass with issues	All four returned destinations were in Florida and below budget. Several itineraries were disproportionately long for the four-night trip, and specialized lodging was presented without recognizing audience restrictions.
S5	Warm beach and nature trip excluding nightlife and major cities	Fail on exclusions	Positive preferences were recognized, but negative constraints were not reliably enforced. Nightlife-oriented destinations remained, and Miami was returned despite the request to avoid major cities.
S6	International trip with safety identified as the highest priority	Partial	Safety scores and advisory information affected scoring, but priority language did not reliably dominate ranking. San Juan was also returned despite the international-only request.
S7	Strict $1,000 total budget with direct-flight preference	Pass with issues	All five recommendations remained under the hard budget ceiling, while the direct-flight preference had limited influence on final ranking and some itinerary/lodging concerns remained.
S8	Seven-night trip restricted exclusively to Japan	Pass	All five returned destinations were in Japan and below the $3,500 budget.
S9	Caribbean-only beach trip, $2,500 budget, and maximum Level 1 advisory requirement	Partial	Geography and budget constraints were satisfied, but a destination with unavailable advisory evidence remained eligible, and Aruba was paired with lodging in Memphis, Tennessee.

Phuket Old Town, and Hola Hostal in Barcelona entered recommendation results despite the intended hostel exclusion. This suggests that the current combination of provider-supplied property classification and name-based filtering is insufficient for reliable accommodation-type validation.

6.6 Flight Practicality

Live flight retrieval generally produced usable route, date, price, connection, and duration information. The evaluation nevertheless exposed a distinction between itinerary availability and itinerary practicality.

Several technically valid itineraries consumed a large fraction of the requested vacation duration. Examples appeared for Naples, Phuket, Athens, Kyoto, and Hilton Head Island. In Scenario S8, Kyoto ranked first despite an outbound itinerary of approximately 20 hours and a return itinerary exceeding 29 hours, including a return layover approaching 14 hours.

These cases indicate that flight practicality is represented in the output more clearly than it is enforced in final ranking. A future ranking component could explicitly penalize the proportion of trip time consumed by transportation.

6.7 Live-Data Availability and Traceability

The evaluated final recommendations generally contained live SerpAPI flight and lodging information rather than local fallback estimates. Structured records preserved source, dates, prices, selected properties, fallback indicators, seasonal-climate evidence, advisory information, preliminary scores, embedding similarity, final scores, and scoring reasons.

This traceability made it possible to localize failures. For example, the Aruba error was a lodging-validation problem rather than a Caribbean candidate-filtering problem: Aruba itself satisfied the geographic request, while its selected lodging did not.

6.8 Evaluation Summary

The scenario evaluation illustrates reliable behavior for several classes of explicit structured input. Numeric budgets, trip durations, months, positive travel categories, and supported geographic restrictions were the strongest recorded behaviors.

The tests also identified limitations that source-code inspection alone would not fully reveal. Negative preference language, relative-priority statements, unavailable advisory information, external lodging geography, accommodation classification, and extreme flight itineraries were handled less reliably. The evaluation therefore supports claims about implemented functionality and observed behavior, but it does not establish objective recommendation reliability or advantage over another recommender.

7 Discussion and Limitations

The evaluation illustrates both the value and limitations of decomposing travel recommendation into separate stages. Explicit filtering constrains the candidate set, semantic embeddings contribute relevance, deterministic scoring evaluates applicable factors, external services provide changing travel information, and the explanation component communicates the resulting evidence. This separation makes individual decisions inspectable but does not prevent errors from propagating from an external-data or parsing stage into final output.

7.1 Rule-Based Language Interpretation

The parser is effective when relevant information is expressed in forms represented by its keyword and regular-expression rules. The evaluation showed reliable behavior for numeric budgets, trip lengths, months, positive travel styles, and several explicit geographic constraints. More intricate semantic expressions were less reliable.

Exclusions such as avoiding nightlife and major cities were not consistently converted into candidate-removal rules. Relative preference statements such as identifying safety as the highest priority or preferring direct flights also did not consistently alter final weighting. Future parsing could represent positive constraints, negative constraints, hard requirements, and soft preferences as distinct objects rather than mapping most detected concepts as positive scoring signals.

7.2 Heuristic Ranking Weights

The final ranking is controlled by developer-defined rewards and penalties rather than values learned from user behavior or extracted through a formal multi-criteria weighting procedure. This grants transparency because each contribution can be traced to the implementation and output, but it limits adaptivity.

A difference of several scoring points should therefore be interpreted as the result of the implemented decision rules rather than as a calibrated probability that one destination is objectively better than another. Future evaluation could perform weight-sensitivity analysis or compare fixed weighting with learned ranking once appropriate training or relevance data exist.

7.3 Static Destination Attributes

The inventory contains stored safety scores, attraction scores, and fallback nightly-cost estimates whose original provenance could not be fully reconstructed; these values should therefore be treated as prototype ranking attributes rather than validated real-time measurements. Travel advisories provide a separate dynamic safety signal retrieved at request time and should not be interpreted as equivalent to the static stored safety score.

7.4 External Data Quality

The scenario tests indicate that external data retrieval and external data validity are separate concerns. Live lodging records were available, but geographically incorrect properties were observed for Jackson and Aruba. Similar issues occurred with accommodation classification when hostel-style properties were represented broadly enough to survive local filtering.

A stronger implementation should validate property coordinates, administrative area, or distance from the intended destination before incorporating lodging price into trip cost. Accommodation type could also be validated using broader property-name patterns and additional metadata rather than relying on the current limited combination of provider classification and name-based checks.

7.5 Flight Practicality

The flight service records durations, connections, layovers, and a practicality category, but the evaluation showed that very long itineraries can remain viable when other destination characteristics are strong. This is especially important for short trips because long travel time can materially reduce usable vacation time.

7.6 Explanation Quality

The language-generation component produces explanations from structured recommendation evidence, but no human-subject evaluation was conducted to measure explanation correctness, trust, usefulness, or comprehension. The existence of an explanation should therefore not be interpreted as evidence that it improves traveler decisions.

The tests also showed that explanations can expose conflicts that the ranking pipeline failed to resolve, which makes them useful for diagnosing recommendation behavior even though they do not correct the ranking itself.

7.7 Evaluation Scope

The evaluation validates observed system functioning rather than recommendation superiority. No labeled relevance dataset was available to establish an objectively correct ranking, and no baseline recommender was evaluated under identical conditions. The results therefore do not support claims that the system is more accurate than alternative recommenders or that its present scoring weights are optimal.

No controlled study of usability, latency, or scalability was conducted. The results should be interpreted as an empirical functional evaluation of an implemented prototype.

8 Future Work

The most immediate improvements follow directly from the scenario findings. Negative preferences and hard semantic constraints should be represented explicitly during request parsing. Advisory requirements should distinguish unavailable evidence from a verified low-risk state. Lodging candidates should undergo geographic validation before contributing to trip cost or viability, and accommodation classification should use stronger metadata and name-based validation than the current filtering rules.

Flight ranking could incorporate an explicit penalty based on total transit time relative to trip duration. Heuristic scoring weights could be subjected to sensitivity analysis or tuned against judged recommendation cases. Persistent recommendation-job storage would improve reliability across backend restarts, and external-service caching could be integrated more consistently.

Longer-term evaluation should use independently judged destination relevance or user studies to assess ranking quality and explanation usefulness. If sufficient interaction data become available, learned ranking methods could be compared with the current deterministic approach while preserving separation between ranking computation and generative explanation.

9 Conclusion

This work presents a travel decision-support system that converts natural-language trip requirements into ranked destination recommendations through a two-stage computational pipeline. Explicitly recognized constraints restrict the candidate set, asymmetric semantic embeddings and deterministic local scoring produce a preliminary ranking, and a limited shortlist receives five-year seasonal

climate, live flight, live lodging, and travel-advisory enrichment. A second deterministic scoring stage is followed by live-data viability filtering before final recommendations are presented.

The implementation separates recommendation computation from natural-language presentation. Gemini receives the completed ranking and supporting evidence only after destination selection, while deterministic explanation generation remains available if the generative service fails. The application also supports persistent destination storage, destination discovery, embedding caching, and candidate replenishment when too few viable results survive live-data validation.

Nine scenario-based tests produced 44 final recommendations. The strongest recorded behaviors involved explicit numerical budgets, dates, positive travel categories, and supported geographic restrictions. The same evaluation exposed weaknesses in negative-preference interpretation, preference-priority handling, missing advisory states, lodging geographic validation, accommodation classification, and extreme flight itineraries. The project therefore demonstrates an implemented and inspectable hybrid recommendation architecture while additionally identifying concrete engineering and evaluation requirements for improving recommendation quality.

References

[1] Mohamed Badouch and Mehdi Boutaounte. 2023. Personalized Travel Recommendation Systems: A Study of Machine Learning Approaches in Tourism. *Journal of Artificial Intelligence, Machine Learning and Neural Network* 3, 3 (2023), 35–45. https://doi.org/10.55529/jaimlnn.33.35.45

[2] JunHo Yoon and Chang Choi. 2023. Real-Time Context-Aware Recommendation System for Tourism. *Sensors* 23, 7 (2023), 3679. https://doi.org/10.3390/s23073679

[3] Xiang Huang. 2022. Personalized Travel Route Recommendation Model of Intelligent Service Robot Using Deep Learning in Big Data Environment. *Journal of Robotics* 2022 (2022), Article 7778592. https://doi.org/10.1155/2022/7778592

[4] Elis Hernawati, Siska Komala Sari, and Dedy Rahman Wijaya. 2021. Combination of Analytic Hierarchy Process and Simple Additive Weighting for Tourist Attractions Recommendation System. *International Journal of Applied Information Technology* 5, 2 (2021), 123–131. https://doi.org/10.25124/ijait.v5i02.4472

[5] Patrizia Beraldi, Annarita De Maio, Filomena Olivito, Giuseppe Potrino, Immacolata Straface, and Antonio Violi. 2021. A Decision Support System for Trip Tourism Recommendation. *International Journal of Transport Development and Integration* 5, 1 (2021), 69–80. https://doi.org/10.2495/TDI-V5-N1-69-80

[6] Yao Rong, Tobias Leemann, Thai-Trang Nguyen, Lisa Fiedler, Peizhu Qian, Vaibhav Unhelkar, Tina Seidel, Gjergji Kasneci, and Enkelejda Kasneci. 2024. Towards Human-Centered Explainable AI: A Survey of User Studies for Model Explanations. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46, 4 (2024), 2104–2119. https://doi.org/10.1109/TPAMI.2023.3331846

[7] Katira Madhuri Kiritbhai and Ripal Ranpara. 2024. A Comprehensive Comparative Analysis of Artificial Intelligence-Based Recommender System Algorithms for Enhancing Travel Search Experience. In *Deep Learning and Visual Artificial Intelligence*, Vishal Goar, Aditi Sharma, Jungpil Shin, and M. Firoz Mridha (Eds.). Algorithms for Intelligent Systems. Springer Nature Singapore, 215–224. https://doi.org/10.1007/978-981-97-4533-3$_1$7